

ALBIN KARINTHOLIL ROBERT

AI/ML Computational Science Associate | Python Application Development & Data Science

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PROFESSIONAL SUMMARY

Python-proficient Data Scientist and former full-stack Application Developer with an MSc in Data Science and Analytics (Distinction) and over two years of hands-on Python experience across data science and applied ML/GenAI work, backed by earlier professional software development experience (Java, HTML, JavaScript, React) at TCS. Comfortable working independently, troubleshooting and debugging applications and data pipelines down to root cause, working with SQL and relational databases, and using Git/GitHub for version control in Agile/Scrum teams. Self-directed experience building REST APIs (Flask, FastAPI) to integrate ML models into applications, alongside strong documentation habits for both technical and non-technical audiences. Combines solid software-engineering fundamentals with data handling, SQL, and applied AI/ML skills, and a track record of collaborating across cross-functional teams to turn requirements into working solutions.

CORE COMPETENCIES

Python Application Development • REST API Development (Flask, FastAPI) • SQL & Database Management • Version Control (Git/GitHub) • Debugging & Troubleshooting • Agile/Scrum • Documentation & Specifications • Machine Learning & Deep Learning • Generative AI / LLMs • Retrieval-Augmented Generation (RAG) • Data Cleaning, Preprocessing & Feature Engineering • Model Evaluation & Explainability (SHAP/LIME) • Data Visualization • Client Communication & Cross-Functional Collaboration

TECHNICAL SKILLS

Programming Languages:

Python, SQL, Java, JavaScript, HTML, CSS

Application Development & APIs:

Flask, FastAPI, REST API Development, Streamlit

Database & Data Engineering:

SQL (MySQL Workbench), Pandas, NumPy, ETL, Feature Extraction, Data Sampling

Version Control & Dev Tools:

Git/GitHub, Jupyter Notebook, Jira, Trello, Agile/Scrum

AI / GenAI:

Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Embeddings (Sentence-Transformers), ChromaDB (Vector DB), LangChain, HuggingFace Transformers (DistilBERT), Prompt Engineering, NLP

ML / Deep Learning:

TensorFlow, Keras, PyTorch, scikit-learn, ANN/CNN, Computer Vision, Transfer Learning (MobileNetV2), XGBoost, Random Forest, SVM, Logistic Regression, Linear Regression, Decision Trees, KNN, Naive Bayes, K-Means, DBSCAN, Hierarchical Clustering, PCA, Apriori (Association Rule Mining), Collaborative & Content-Based Filtering

Time-Series Forecasting:

ARIMA, Exponential Smoothing, Moving Average, Prophet, ADF Stationarity Testing, ACF/PACF Analysis

Model Evaluation & Visualization:

Accuracy, R^2 , RMSE, MAE, ROC-AUC, Precision/Recall, Cross-Validation, Confusion Matrix, Matplotlib, Seaborn, Power BI

PROFESSIONAL EXPERIENCE

Data Science Trainee | Techmindz | Feb 2026 - Present | Infopark, Kochi, India (Full-Time)

- Practiced regression modeling using Linear Regression, Decision Tree, KNN, and Random Forest across multiple datasets (vehicles, concrete, housing, insurance); built and evaluated a Linear Regression car price prediction model achieving an R^2 of 0.45 (RMSE 11,339, MAE 7,359), and derived business recommendations on key pricing drivers (vehicle age, mileage, manufacturer, condition).
- Built time-series forecasting models (ARIMA, Exponential Smoothing, Moving Average, Prophet) on a year of daily sales data, applying ADF stationarity testing and ACF/PACF plots to guide model parameter selection.
- Applied unsupervised learning techniques for business use cases: K-Means, DBSCAN, and Hierarchical Clustering for customer segmentation (including a structured hierarchical clustering project on credit card data), PCA for dimensionality reduction (Iris, Wine Quality datasets), and the Apriori algorithm for market basket analysis and association rule mining.
- Currently building “Career Intelligence AI,” an end-to-end Generative AI application: fine-tuning a HuggingFace DistilBERT transformer for text classification and designing a Retrieval-Augmented Generation (RAG) pipeline (embeddings, ChromaDB, LangChain) for a conversational career assistant grounded in real job posting data.
- Diagnosed and corrected a data-leakage issue in an earlier classification project by tracing unexpected model behaviour back to its root cause, then rebuilding the pipeline logic to eliminate it — practicing the same debugging and troubleshooting discipline applied to application code.

Data Scientist | Technocolabs | Jul 2022 - Apr 2023 | Remote, India

- Start-up Investment Analysis: collected, cleaned, and preprocessed financial datasets to support a client's investment decision-making, applying end-to-end EDA to surface key patterns and trends.
- Developed an interactive real-time dashboard visualizing key financial metrics, enabling data-driven monitoring of startup performance.
- Communicated findings through structured presentations to non-technical stakeholders, translating complex data insights into actionable investment recommendations.
- Peer-to-Peer Lending Default Prediction: designed and built a classification model achieving 80%+ accuracy in identifying potential loan defaulters, directly reducing financial risk.
- Led data preprocessing and feature selection using Python (Pandas, scikit-learn); performed data cleaning, encoding, and outlier handling.
- Mortgage-Backed Securities Prepayment Risk: conducted comprehensive EDA using Pandas, Matplotlib, and Seaborn; built and validated a multi-model classification pipeline (Logistic Regression, SVM, Gaussian Discriminant Analysis) and identified the optimal model for deployment.

Systems Engineer | Tata Consultancy Services (TCS) | Jun 2018 - Apr 2021 | Kochi, India

- Contributed to agile application development projects using Jira, Trello, and Git/GitHub for workflow and version control management, including troubleshooting and debugging issues during development and testing.
- Developed and maintained front-end components using HTML, JavaScript, CSS, and React as part of a collaborative cross-functional team.
- Participated in daily scrum ceremonies, demonstrating strong communication and remote collaboration skills.

RECENT PROJECTS & CONTINUOUS LEARNING

Telecom Customer Churn Prediction | End-to-End CRISP-DM ML Project

- Built a full CRISP-DM pipeline on the IBM Telco Customer Churn dataset, training and comparing 10 classification models.
- Identified and corrected a SMOTE data-leakage issue in the cross-validation setup, rebuilding the pipeline to apply resampling only within each training fold.
- Confirmed Logistic Regression as the best-performing, most robust model on unseen data (ROC-AUC 0.8465); applied SHAP and LIME for explainability to surface key churn drivers for business stakeholders.

Dog Breed Image Classifier | CNN with Transfer Learning (TensorFlow/Keras)

- Built and debugged an end-to-end image classification pipeline in TensorFlow/Keras, diagnosing and fixing a broken data pipeline before model training.
- Applied transfer learning with MobileNetV2 to classify images across 12 dog breed classes, reaching ~85% validation accuracy.
- Worked hands-on with data augmentation, feature extraction from pretrained convolutional layers, and model evaluation on a held-out validation set.

Data Engineering & Cloud Fundamentals | Coursera / IBM

- Completed SQL and relational database design coursework via an IBM Data Engineering lab; completed hands-on exercises locally in MySQL Workbench.

ACADEMIC PROJECT

Mental Health Illness Prediction Model | University of Hertfordshire — MSc Final Project

- Built an end-to-end ML pipeline to predict mental health illness from a Kaggle survey dataset (mental health conditions, education, employment, hospitalization history, demographics), including missing-value handling, label encoding, SelectKBest feature selection, SMOTE class balancing, and StandardScaler normalization.
- Designed, trained, and evaluated five classification algorithms — Neural Networks, Random Forest, Decision Tree, Naive Bayes, and KNN — comparing performance across accuracy, precision, recall, F1-score, and AUC.
- Achieved the best result with a Neural Network model (98.76% accuracy, 0.08% false positive rate), outperforming KNN (90.20%), Random Forest (89.22%), Decision Tree (85.29%), and Naive Bayes (78.43%).
- Built a Python Tkinter GUI to serve model predictions and display performance evaluation results to end users.

EDUCATION

MSc, Data Science and Analytics | University of Hertfordshire | Sep 2021 - Sep 2023 | Hatfield, UK | Grade: Distinction
 Relevant Modules: Data Science & Analytics, Data Mining, Machine Learning & Neural Networks, Statistical Foundations, Legal & Ethical Practices

BEng, Mechanical Engineering | VSB Engineering College | Jun 2014 - May 2018 | Karur, India | CGPA: 8.41/10